gave Penzias and Wilson, thirty-five years later, to find microwave noise in their receiver, as we saw in chapter 3, which led to the discovery of the cosmic microwave background.

By spending a couple of years painstakingly tracking and timing the static hiss that registered on his jury-rigged antenna, Jansky had discovered that radio waves emanate not just from local thunderstorms and other known terrestrial sources, but also from the center of the Milky Way galaxy. That region of the sky swung by the telescope's field of view every twenty-three hours and fifty-six minutes: exactly the period of Earth's rotation in space and thus exactly the time needed to return the galactic center to the same angle and elevation on the sky. Karl Jansky published his results under the title "Electrical Disturbances Apparently of Extraterrestrial Origin."†††††

With that observation, radio astronomy was born—but minus Jansky himself. Bell Labs retasked him, preventing him from pursuing the fruits of his own seminal discovery. A few years later, though, a self-starting American named Grote Reber, from Wheaton, Illinois, built a thirty-foot-wide, metal-dish radio telescope in his own backyard. In 1938, under nobody's employ, Reber confirmed Jansky's discovery, and spent the next five years making low-resolution maps of the radio sky.

Reber's telescope, though without precedent, was small and crude by today's standards. Modern radio telescopes are quite another matter. Unbound by backyards, they're sometimes downright humongous. MK 1, which began its working life in 1957, is the planet's first genuinely gigantic radio telescope—a single, steerable, 250-foot-wide, solid-steel dish at the Jodrell Bank Observatory near Manchester, England. A couple of months after MK 1 opened for business, the Soviet Union launched *Sputnik 1*, and Jodrell Bank's dish suddenly became just the thing to track the little orbiting hunk of hardware—making it the forerunner of today's Deep Space Network for tracking planetary space probes.

The world's largest radio telescope, completed in 2016, is called the Five-hundred-meter Aperture Spherical radio Telescope, or "FAST" for short. It was built by China in their Guizhou Province, and is larger in area than thirty football fields. If aliens ever give us a call, the Chinese will be the first to know.

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